

Decoding Breast Cancer Morphology with Machine Learning

Supervised & Unsupervised Learning on the Breast Cancer Wisconsin Dataset

UM5BM753 • Group no. - 7

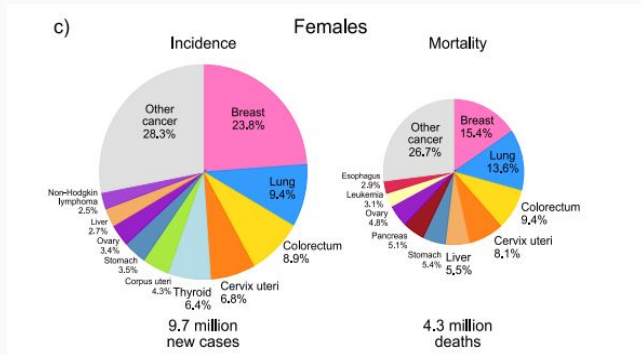
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Breast Cancer: global burden and the need for automated diagnostics

Breast cancer is one of the most prevalent malignancies affecting women worldwide and is a leading cause of cancer-related morbidity.

According to the latest global estimates, breast cancer accounted for approximately 2.3 million new cases worldwide in 2022 of all newly diagnosed cancers, and was responsible for approximately 670,000 deaths

MIR 0.0 0.2 0.4 0.6 0.8 1.0
Higher MIR suggests poorer survival / late diagnosis



- The Survival Gap: Early detection improves survival rates.
- The standard Fine-Needle Aspiration (FNA) biopsy relies on visual review by pathologists, which is time-consuming and vulnerable to subjective human error.
- The Computational Need: There is a clinical need for automated tools to objectively quantify cellular morphology to accelerate and standardize diagnosis.

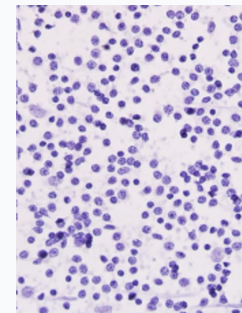
Inside the Data: The Breast Cancer Wisconsin Dataset

569 SAMPLES
357 benign • 212
malignant

30 FEATURES
Continuous nuclear
morphology

10×3
Mean • SE • Worst

RADIUS • TEXTURE • PERIMETER • AREA • SMOOTHNESS
COMPACTNESS • CONCAVITY • CONCAVE POINTS • SYMMETRY • FRACTAL
DIMENSION



FNA cytology • cell
nuclei

Project Aim

Q1.

Can machine learning algorithms accurately **predict** whether a tumor is **benign** or **malignant** based on cellular **morphology**?

Methods:
Logistic Regression, SVM-RBF, and Random Forest.

Q2.

Do the **morphological features** naturally **separate** the samples into distinct **subpopulations** that align with the **clinical diagnostic** classes?

Methods:
K-means, Hierarchical clustering, and Gaussian Mixture Models (GMM).

Q3.

What are the **key features** driving the separation between benign and malignant samples?

Methods:
Principal Component Analysis (PCA) and feature selection.

From Raw Data to Model-Ready Data

1 RAW DATA
 569×32
357 B • 212 M

2 QUALITY
Missing = 0
Duplicates = 0
Infinite = 0

3 PREPARE
Remove ID
B = 0 • M = 1
30 features

4 SPLIT
80% = 455
20% = 114
Stratified • seed 42

5 MODEL
LR • SVM • RF
K-means • HC •
GMM

OUTLIER CHECK • $1.5 \times \text{IQR}$ cutoff • 608 extreme feature values flagged • RETAINED as potentially real biological variation
SAME supervised split used across models → fair comparison

BARAA

Teamwork

Baraa

SUPERVISED

Logistic Regression

UNSUPERVISED

k-means

Sara

SUPERVISED

SVM-RBF

UNSUPERVISED

Hierarchical clustering

Nikita

SUPERVISED

Random Forest

UNSUPERVISED

Gaussian Mixture Model

Prepare the data

- Checked quality: no missing values, no duplicates
- Reviewed outliers and kept them as real biological variation
- Created one shared 80/20 stratified split, used by all models

Compare the models

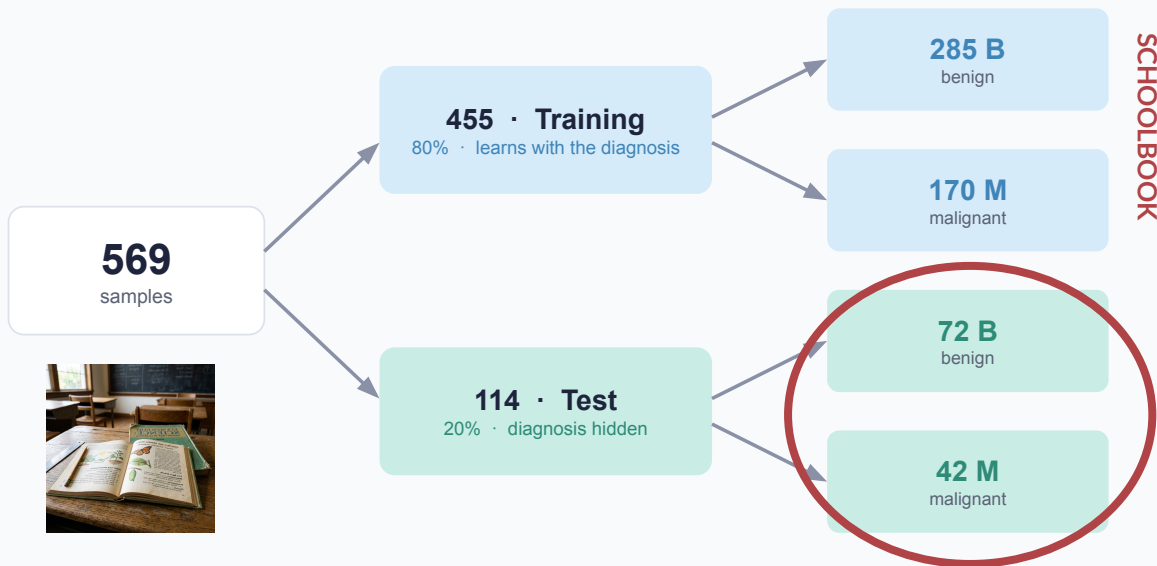
- Same test set and same metrics for every model
- Supervised: ROC-AUC, sensitivity, specificity, F1
- Unsupervised: ARI, NMI, silhouette

Interpret the results

- Identified the most discriminative nuclear features
- Compared mean, SE and worst measurements
- Linked the findings to tumor biology

Supervised Models

Learning a mapping from features to a **known label** on training data, then evaluating generalization on unseen test data.



Stratified split: both sets keep the same mix ($\approx 63\%$ benign · 37% malignant) · `random_state = 42`

**CORRECT
ANSWER
'UNSEEN'**

PERFORMERS

Logistic Regression

a weighted risk score, turned into a probability

Support Vector Machine (SVM)

the widest possible border between the two groups

Random Forest

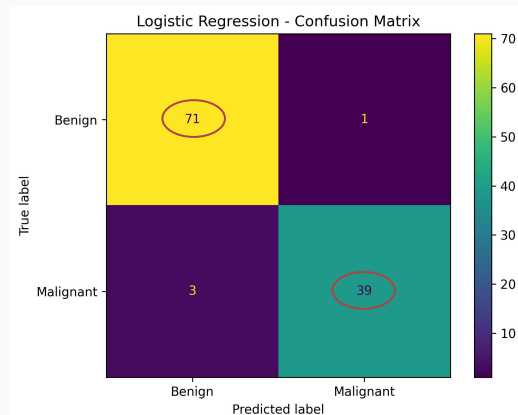
hundreds of decision trees that vote

~97% of unseen samples classified correctly

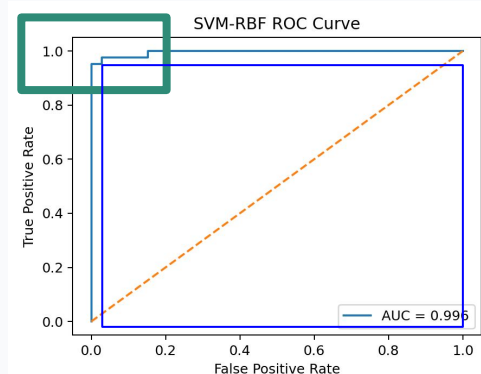
The only difference: one benign sample (a false alarm by logistic regression). Every model missed 3 of 42 cancers.



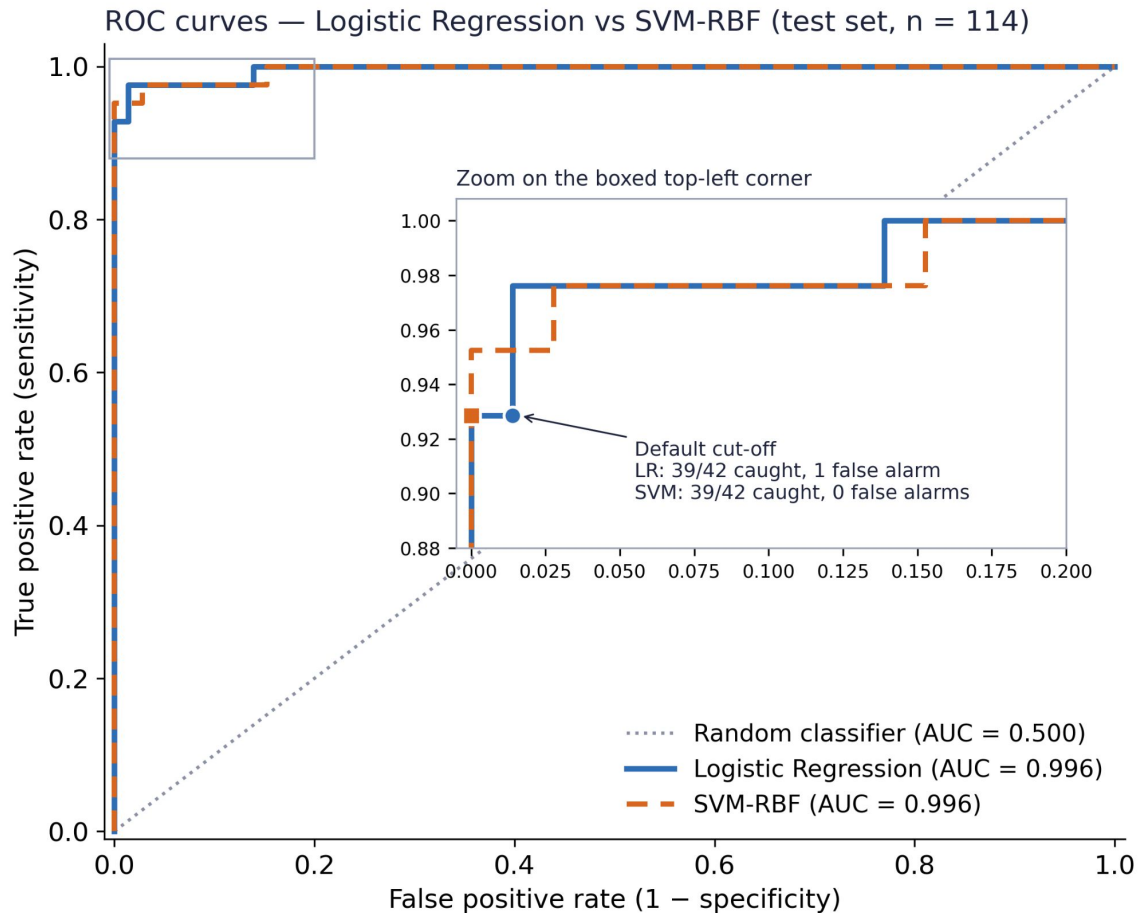
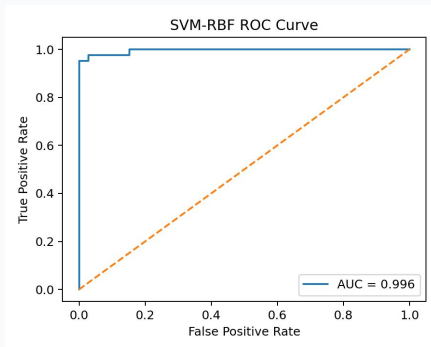
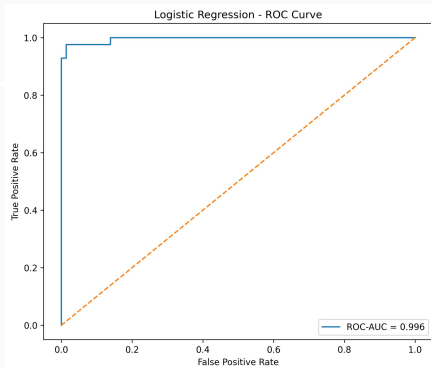
Model	ROC-AUC	Sensitivity	Specificity	F1	Accuracy	Missed cancers	False alarms
Logistic Regression	0.996	92.9% (39/42)	98.6% (71/72)	0.951	96.5%	3	1
SVM-RBF	0.996	92.9% (39/42)	100% (72/72)	0.963	97.4%	3	0
Random Forest	0.997	92.9% (39/42)	100% (72/72)	0.963	97.4%	3	0



Metric	Formula / definition	Question it answers
ROC-AUC Receiver Operating Characteristic – Area Under the Curve	Area under the curve of TPR (sensitivity) vs FPR (1 – specificity) across all thresholds 0.5 = chance · 1 = perfect	How well does the model separate malignant from benign?
Sensitivity recall · true positive rate	$TP / (TP + FN)$	Of all malignant masses, how many were detected?
Specificity true negative rate	$TN / (TN + FP)$	Of all benign masses, how many were correctly classified?
F1 score	$2 \cdot \text{Precision} \cdot \text{Sensitivity} / (\text{Precision} + \text{Sensitivity})$	How well does the model <i>balance</i> correct detection against false alarms?

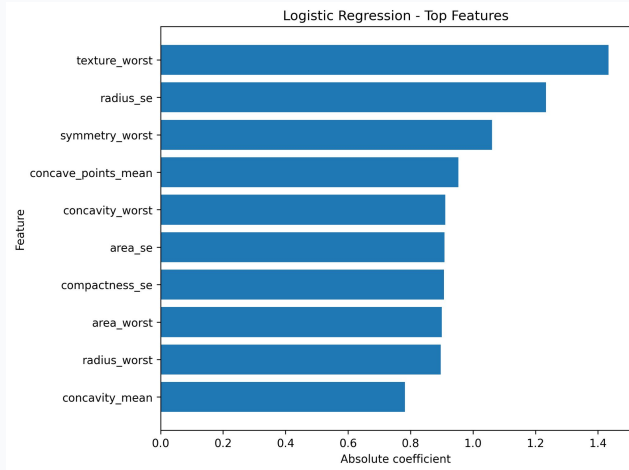


ANNEX

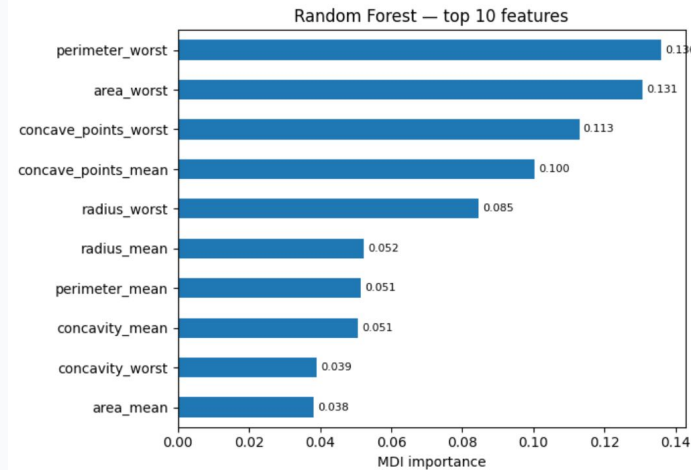


Annex — Which features drive the predictions?

Logistic Regression



Random Forest



In both top-10 lists

concave points (mean)
concavity (mean)
concavity (worst)
radius (worst)
area (worst)

Same biological signal. The two models rank features differently, but both put the same two kinds of features at the top:

- **Size of the nucleus:** radius, area, perimeter. Malignant nuclei are bigger.
- **Shape of the border:** concavity and concave points, meaning dents and indentations in the nuclear border. Malignant nuclei have irregular, notched borders.

Supervised learning — what we conclude

Nuclear morphology predicts the diagnosis. All three models reached $AUC \approx 0.996$ and $\sim 97\%$ accuracy on 114 unseen samples — and each missed the same number of cancers (3 of 42).

Our questions → answers

Q1 • Predict: yes. AUC 0.996–0.997; sensitivity 92.9%; specificity 98.6–100%.

Q3 • Interpret: the signal is **nuclear size** (area, radius, perimeter) and **border irregularity** (concavity, concave points).

Complexity not needed: linear LR $\approx$ SVM and RF (one sample apart) → classes almost linearly separable.

→ **Q2:** errors lie where benign and malignant morphology overlap — tested without labels next.

Consistent with literature

Original study (Street et al., 1993): one linear plane on 3 features (worst area, worst smoothness, mean texture) separated the classes; the Xcyt system then correctly diagnosed 176 new patients.

Recent benchmark (Cheng & Yu, 2025, preprint): LR 98.2% accuracy, AUC 0.995; complex models added little.

Cytopathology: enlarged, pleomorphic nuclei with irregular contours = classic malignancy criteria.

Human FNA reading: sensitivity 74%, specificity 96% (Wang et al., 2017) — not directly comparable.

Limits → next steps

Small test set: 1 cancer = 2.4 points of sensitivity; model differences are within noise.

Single split, one historical dataset → cross-validation and external validation.

3 missed cancers = costliest error → lower the cut-off (set on training data) to favour sensitivity.

Correlated features → read feature families, not single ranks.

Take-home: nuclear morphology gives an accurate, interpretable and biologically coherent prediction of malignancy — a decision-support tool for the cytopathologist, not a replacement.

UNSUPERVISED LEARNING

Each method asks a different question of the same 30 standardized morphology features.

K-means

Which samples sit closest to the same centroid?

Choose k → centroids → nearest points → update centroids → repeat

$k = 2$ selected by silhouette score

Hierarchical

Which samples merge together step by step?

Each point alone → merge closest groups → repeat → dendrogram → cut

Dendrogram cut at 2 clusters for comparison

GMM

Which probability distribution most likely generated each sample?

Start distributions → calculate probabilities → update distributions → repeat .

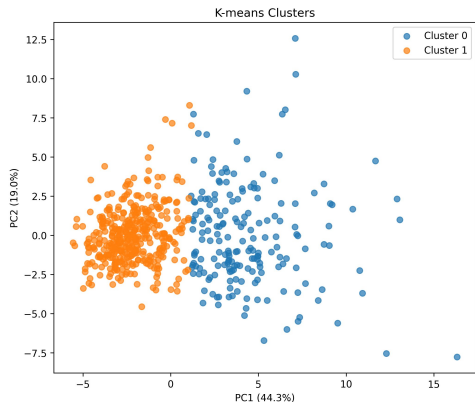
2 components used to compare with B / M

Important: diagnosis labels are never used to build the clusters — they are used only afterwards to measure how well the discovered groups align with diagnosis.

What the three clustering methods actually found

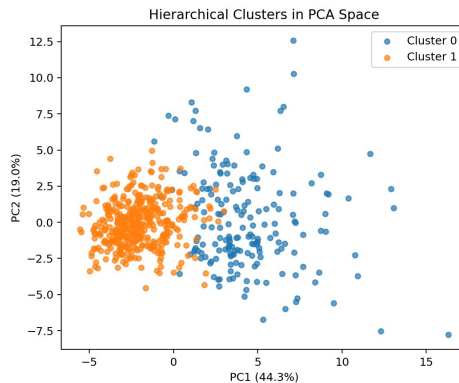
Real PCA projections from the executed analyses. PCA is used only to visualize the cluster assignments in two dimensions.

K-means



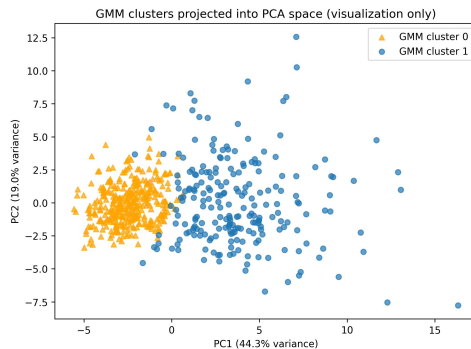
compact centroid groups

Hierarchical



nested structure, more overlap

GMM



probabilistic separation

The pictures suggest structure; the next slide quantifies it with ARI, NMI and silhouette.

Which method aligns best with diagnosis?

No single metric tells the whole story: ARI/NMI measure agreement with diagnosis; silhouette measures internal cluster separation.

Method	ARI	NMI	Silhouette	Best at
K-means	0.6707	0.5546	0.3450	Silhouette
Hierarchical	0.5750	0.4569	0.3394	—
GMM	0.6779	0.5603	0.3157	ARI + NMI

ARI

Agreement with diagnosis, corrected
for chance

0 = random • 1 = perfect

NMI

Shared information between cluster
and diagnosis labels

0 = none • 1 = perfect

Silhouette

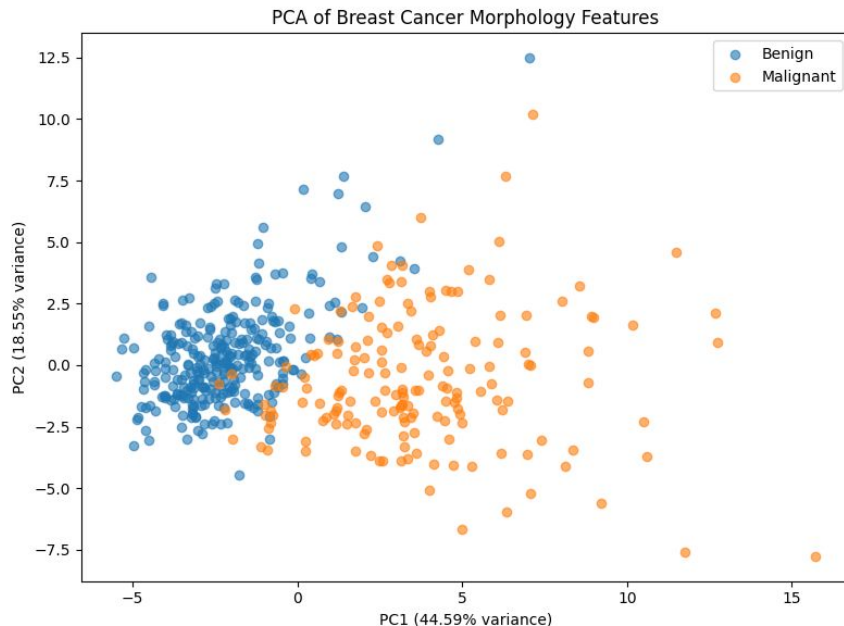
How compact and separated the
clusters are

higher = cleaner geometry

GMM has the strongest diagnosis agreement (ARI/NMI); K-means has the cleanest internal separation (silhouette).

PCA shows the same morphology signal in a smaller space

This is the actual PCA output from our feature-selection notebook. PCA itself is unsupervised; diagnosis is used only to color the points.



PC1
44.59%

PC2
18.55%

7 principal components
retain 91.13% of total variance

Reduced representations still perform strongly

Worst 10: 97.89% $\pm$ 1.19% CV accuracy
PCA 7: ROC-AUC 0.9946

Interpretation: benign samples concentrate toward negative PC1, while malignant samples extend toward positive PC1 — with overlap rather than a perfect boundary.

Conclusion

Yes: nuclear morphology predicts diagnosis, reveals related structure even without labels, and much of the signal survives feature reduction.

WITH LABELS

PREDICTION WORKS

All three supervised models strongly distinguish benign from malignant morphology.

WITHOUT LABELS

STRUCTURE REMAINS

Natural clusters still align with diagnosis, but overlap reveals biological heterogeneity.

WITH FEWER FEATURES

MOST SIGNAL SURVIVES

Worst-10 features and a 7-PC representation preserve most of the useful information.

Malignant and benign samples occupy different — but overlapping — regions of morphology space. Across methods, the recurring signal is nuclear size and border irregularity, especially concavity and concave points.

LIMITATIONS • 569 samples • correlated features • internal validation only • no external clinical cohort → strong dataset evidence, not clinical validation.

Thank you — questions?

Code, notebooks, figures and results are fully reproducible from the project resources below.

<https://github.com/nikitameena305/breast-cancer-morphology-ml>

https://drive.google.com/drive/folders/1Kl0SP20VG_jlyq0xev1KHskRkCNmTwvg